

CHAPTER 3

ANALYSIS OF FOOD COMPONENTS

Proximate analysis of foods refers to determining the major components of foods like moisture, ash (total minerals), lipids, protein, and carbohydrates. Other minor components of foods are also analysed. The performance of many analytical methods is affected by the food matrix (i.e., its major chemical components, especially lipid, protein, and carbohydrate). In food analysis, it is usually the food matrix that presents the greatest challenge to the analysis. For example, high-fat or high-sugar foods can cause different types of interferences than low-fat or low-sugar foods. Digestion procedures and extraction steps necessary for accurate analytical results can be very dependent on the food matrix. The complexity of various food systems often requires having not just one technique available for a specific food component, but multiple techniques and procedures, as well as the knowledge about which to apply to a specific food matrix. Analysis of food components will be discussed below.

3.1 Determination of Moisture and Total Solids

Moisture content is one of the most commonly measured properties of food materials. It is important to food scientists for a number of different reasons:

- ➡ **Legal and Labeling Requirements:** There are legal limits to the maximum or minimum amount of water that must be present in certain types of food.
- ➡ **Economic:** The cost of many foods depends on the amount of water they contain - water is an inexpensive ingredient, and manufacturers often try to incorporate as much as possible in a food, without exceeding some maximum legal requirement.
- ➡ **Microbial Stability:** The propensity of microorganisms to grow in foods depends on their water content. For this reason many foods are dried below some critical moisture content.
- ➡ **Food Quality:** The texture, taste, appearance and stability of foods depend on the amount of water they contain.
- ➡ **Food Processing Operations:** Knowledge of the moisture content is often necessary to predict the behavior of foods during processing, *e.g.* mixing, drying, flow through a pipe or packaging.

It is therefore important for food scientists to be able to reliably measure moisture contents. A number of analytical techniques have been developed for this purpose, which vary in their

accuracy, cost, speed, sensitivity, specificity, ease of operation, *etc.* The choice of an analytical procedure for a particular application depends on the nature of the food being analyzed and the reason the information is needed.

3.1.1 Properties of Water in Foods

The moisture content of a food material is defined through the following equation:

$$\% \text{ Moisture} = (m_w/m_{\text{sample}}) \times 100 \quad (3.1)$$

Where m_w is the mass of the water and m_{sample} is the mass of the sample. The mass of water is related to the number of water molecules (n_w) by the following expression:

$$m_w = n_w M_w / N_A \quad (3.2)$$

Where M_w is the molecular weight of water (18.0 g per mole) and N_A is Avadagro's number (6.02×10^{23} molecules per mole). In principle, the moisture content of a food can therefore be determined accurately by measuring the number or mass of water molecules present in a known mass of sample. It is not possible to directly measure the number of water molecules present in a sample because of the huge number of molecules involved. A number of analytical techniques commonly used to determine the moisture content of foods are based on determinations of the mass of water present in a known mass of sample. Nevertheless, as we will see later, there are a number of practical problems associated with these techniques that make highly accurate determinations of moisture content difficult or that limit their use for certain applications. For these reasons, a number of other analytical methods have been developed to measure the moisture content of foods that do not rely on direct measurement of the mass of water in a food. Instead, these techniques are based on the fact that the water in a food can be distinguished from the other components in some measurable way.

An appreciation of the principles, advantages and limitations of the various analytical techniques developed to determine the moisture content of foods depends on an understanding of the molecular characteristics of water. A water molecule consists of an oxygen atom covalently bound to two hydrogen atoms (H_2O). Each of the hydrogen atoms has a small positive charge (δ^+), while the oxygen atom has two lone pairs of electrons that each has a small negative charge (δ^-).

Consequently, water molecules are capable of forming relatively strong hydrogen bonds ($\text{O}-\text{H}^{\delta+} \rightarrow \delta-\text{O}$) with four neighboring water molecules. The strength and directionality of these hydrogen bonds are the origin of many of the unique physicochemical properties of water.

The development of analytical techniques to determine the moisture content of foods depends on being able to distinguish water (the "analyte") from the other components in the food (the "matrix"). The characteristics of water that are most commonly used to achieve this are: its relatively low boiling point; its high polarity; its ability to undergo unique chemical reactions with certain reagents; its unique electromagnetic absorption spectra; and, its characteristic physical properties (density, compressibility, electrical conductivity and refractive index).

Despite having the same chemical formula (H_2O) the water molecules in a food may be present in a variety of different molecular environments depending on their interaction with the surrounding molecules. The water molecules in these different environments normally have different physiochemical properties:

- ➡ Bulk water: Bulk water is free from any other constituents, so that each water molecule is surrounded only by other water molecules. It therefore has physicochemical properties that are the same as those of pure water, *e.g.*, melting point, boiling point, density, compressibility, heat of vaporization, electromagnetic absorption spectra.
- ➡ Capillary or trapped water: Capillary water is held in narrow channels between certain food components because of capillary forces. Trapped water is held within spaces within a food that are surrounded by a physical barrier that prevents the water molecules from easily escaping, *e.g.*, an emulsion droplet or a biological cell. The majority of this type of water is involved in normal water-water bonding and so it has physicochemical properties similar to that of bulk water.
- ➡ Physically bound water: A significant fraction of the water molecules in many foods are not completely surrounded by other water molecules, but are in molecular contact with other food constituents, *e.g.* proteins, carbohydrates or minerals. The bonds between water molecules and these constituents are often significantly different from normal water-water bonds and so this type of water has different physicochemical properties than bulk water *e.g.*, melting point, boiling point, density, compressibility, heat of vaporization, electromagnetic absorption spectra.
- ➡ Chemically bound water: Some of the water molecules present in a food may be chemically bonded to other molecules as water of crystallization or as hydrates, *e.g.*

$\text{NaSO}_4 \cdot 10\text{H}_2\text{O}$. These bonds are much stronger than the normal water-water bond and therefore chemically bound water has very different physicochemical properties to bulk water, *e.g.*, lower melting point, higher boiling point, higher density, lower compressibility, higher heat of vaporization, different electromagnetic absorption spectra.

Foods are heterogeneous materials that contain different proportions of chemically bound, physically bound, capillary (trapped) or bulk water. In addition, foods may contain water that is present in different physical states: gas, liquid or solid. The fact that water molecules can exist in a number of different molecular environments, with different physicochemical properties, can be problematic for the food analyst trying to accurately determine the moisture content of foods. Many analytical procedures developed to measure moisture content are more sensitive to water in certain types of molecular environment than to water in other types of molecular environment. This means that the measured value of the moisture content of a particular food may depend on the experimental technique used to carry out the measurement. Sometimes food analysts are interested in determining the amounts of water in specific molecular environments (*e.g.*, physically bound water), rather than the total water content. For example, the rate of microbial growth in a food depends on the amount of bulk water present in a food, and not necessarily on the total amount of water present. There are analytical techniques available that can provide some information about the relative fractions of water in different molecular environments (*e.g.*, NMR, vapor pressure).

3.1.2. Sample preparation

Selection of a representative sample and prevention of changes in the properties of the sample prior to analysis are two major potential sources of error in any food analysis procedure. When determining the moisture content of a food it is important to prevent any loss or gain of water. For this reason, exposure of a sample to the atmosphere, and excessive temperature fluctuations, should be minimized. When samples are stored in containers it is common practice to fill the container to the top to prevent a large headspace, because this reduces changes in the sample due to equilibration with its environment. The most important techniques developed to measure the moisture content of foods are discussed below.

3.1.1. Methods of Food Moisture Analysis

There are different methods to analyse moisture or total solid from food. These methods are illustrated below.

1. Evaporation Methods (Oven drying method)

A. Principles

These methods rely on measuring the mass of water in a known mass of sample. The moisture content is determined by measuring the mass of a food before and after the water is removed by evaporation:

$$\% \text{ moisture} = \frac{M_{\text{dried}}}{M_{\text{initial}}} \times 100 \quad (3.3)$$

Here, M_{INITIAL} and M_{DRIED} are the mass of the sample before and after drying, respectively. The basic principle of this technique is that water has a lower boiling point than the other major components within foods, *e.g.*, lipids, proteins, carbohydrates and minerals.

Sometimes a related parameter, known as the *total solids*, is reported as a measure of the moisture content. The total solids content is a measure of the amount of material remaining after all the water has been evaporated. Thus, $\% \text{Total solids} = (100 - \% \text{Moisture})$. To obtain an accurate measurement of the moisture content or total solids of a food using evaporation methods it is necessary to remove all of the water molecules that were originally present in the food, without changing the mass of the food matrix. This is often extremely difficult to achieve in practice because the high temperatures or long times required to remove all of the water molecules would lead to changes in the mass of the food matrix, *e.g.*, due to volatilization or chemical changes of some components. For this reason, the drying conditions used in evaporation methods are usually standardized in terms of temperature and time so as to obtain results that are as accurate and reproducible as possible given the practical constraints. Using a standard method of sample preparation and analysis helps to minimize sample-to-sample variations within and between laboratories.

B. Evaporation Devices (Instruments used)

The thermal energy used to evaporate the water from a food sample can be provided directly (*e.g.*, transfer of heat from an oven to a food) or indirectly (*e.g.*, conversion of electromagnetic radiation incident upon a food into heat due to absorption of energy by the water molecules).

Convection and forced draft ovens: Weighed samples are placed in an oven for a specified time and temperature (*e.g.* 3 hours at 100 °C) and their dried mass are determined, or they are dried until they reach constant mass. The thermal energy used to evaporate the water is applied directly to the sample *via* the shelf and air that surround it. There are often considerable temperature variations within convection ovens, and so precise measurements are carried out using *forced draft ovens* that circulate the air so as to achieve a more uniform temperature distribution within the oven. Samples that contain significant quantities of carbohydrates that might undergo chemical changes or volatile materials other than water should not be dried in convection or forced draft oven. Many official methods of analysis are based on forced draft ovens. In forced draft ovens the hot air is circulated through the samples by using external forces while in convection oven there is no external force to circulate the hot air.



Figure 3.1 Oven

Vacuum oven: Weighed samples are placed under reduced pressure (typically 25-100 mm Hg) in a vacuum oven for a specified time and temperature and their dried mass is determined. The thermal energy used to evaporate the water is applied directly to the sample *via* the metallic shelf that it sits upon. There is an air inlet and outlet to carry the moisture lost from the sample out of the vacuum oven, which prevents the accumulation of moisture within the oven. The boiling point of water is reduced when it is placed under vacuum. Drying foods in a vacuum oven therefore has a number of advantages over conventional oven drying techniques. If the sample is heated at the same temperature, drying can be carried out much quicker. Alternatively, lower temperatures can be used to remove the moisture (*e.g.* 70°C instead of 100

°C), and so problems associated with degradation of heat labile substances can be reduced. A number of vacuum oven methods are officially recognized.

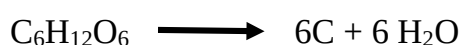
Microwave oven: Weighed samples are placed in a microwave oven for a specified time and power-level and their dried mass is weighed. Alternatively, weighed samples may be dried until they reach constant final mass - analytical microwave ovens containing balances to continuously monitor the weight of a food during drying are commercially available. The water molecules in the food evaporate because they absorb microwave energy, which causes them to become thermally excited. The major advantage of microwave methods over other drying methods is that they are simple to use and rapid to carry out. Nevertheless, care must be taken to standardize the drying procedure and ensure that the microwave energy is applied evenly across the sample. A number of microwave oven drying methods are officially recognized.

Infrared lamp drying: The sample to be analyzed is placed under an infrared lamp and its mass is recorded as a function of time. The water molecules in the food evaporate because they absorb infrared energy, which causes them to become thermally excited. One of the major advantages of infrared drying methods is that moisture contents can be determined rapidly using inexpensive equipment, *e.g.*, 10-25 minutes. This is because the IR energy penetrates into the sample, rather than having to be conducted and convected inwards from the surface of the sample. To obtain reproducible measurements it is important to control the distance between the sample and the IR lamp and the dimensions of the sample. IR drying methods are not officially recognized for moisture content determinations because it is difficult to standardize the procedure. Even so, it is widely used in industry because of its speed and ease of use.

C. Practical Considerations

1. **Sample dimensions:** The rate and extent of moisture removal depends on the size and shape of the sample, and how finely it is ground. The greater the surface area of material exposed to the environment, the faster the rate of moisture removal.
2. **Clumping and surface crust formation:** Some samples tend to clump together or form a semi-permeable surface crust during the drying procedure. This can lead to erroneous and irreproducible results because the loss of moisture is restricted by the clumps or crust. For this reason samples are often mixed with dried sand to prevent clumping and surface crust formation.

3. Elevation of boiling point: Under normal laboratory conditions pure water boils at 100 °C. Nevertheless, if solutes are present in a sample the boiling point of water is elevated. This is because the partial vapor pressure of water is decreased and therefore a higher temperature has to be reached before the vapor pressure of the system equals the atmospheric pressure. Consequently, the rate of moisture loss from the sample is slower than expected. The boiling point of water containing solutes (T_b) is given by the expression, $T_b = T_0 + 0.51m$, where T_0 is the boiling point of pure water and m is the molality of solute in solution (mol/kg of solvent).
4. Water type: The ease at which water is removed from a food by evaporation depends on its interaction with the other components present. Free water is most easily removed from foods by evaporation, whereas more severe conditions are needed to remove chemically or physically bound water. Nevertheless, these more extreme conditions can cause problems due to degradation of other ingredients which interfere with the analysis.
5. Decomposition of other food components: If the temperature of drying is too high, or the drying is carried out for too long, there may be decomposition of some of the heat-sensitive components in the food. This will cause a change in the mass of the food matrix and lead to errors in the moisture content determination. It is therefore normally necessary to use a compromise time and temperature, which are sufficient to remove most of the moisture, but not too long to cause significant thermal decomposition of the food matrix. One example of decomposition that interferes with moisture content determinations is that of carbohydrates.



The water that is released by this reaction is not the water we are trying to measure and would lead to an overestimation of the true moisture content. On the other hand, a number of chemical reactions that occur at elevated temperatures lead to water absorption, *e.g.*, sucrose hydrolysis (sucrose + H₂O $\longrightarrow$ fructose + glucose), and therefore lead to an underestimation of the true moisture content. Foods that are particularly susceptible to thermal decomposition should be analyzed using alternative methods, *e.g.* chemical or physical.

6. Volatilization of other food components: It is often assumed that the weight loss of a food upon heating is entirely due to evaporation of the water. In practice, foods often

contain other volatile constituents that can also be lost during heating, *e.g.*, flavors or odors. For most foods, these volatiles only make up a very small proportion and can therefore be ignored. For foods that do contain significant amounts of volatile components (*e.g.* spices and herbs) it is necessary to use alternative methods to determine their moisture content, *e.g.*, distillation, chemical or physical methods.

7. High moisture samples: Food samples that have high moisture contents are usually dried in two stages to prevent "spattering" of the sample, and accumulation of moisture in the oven. Spattering is the process whereby some of the water jumps out of the food sample during drying, carrying other food constituents with it. For example, most of the moisture in milk is removed by heating on a steam bath prior to completing the drying in an oven.
8. Temperature and power level variations: Most evaporation methods stipulate a definite temperature or power level to dry the sample so as to standardize the procedure and obtain reproducible results. In practice, there are often significant variations in temperatures or power levels within an evaporation instrument, and so the efficiency of the drying procedure depends on the precise location of the sample within the instrument. It is therefore important to carefully design and operate analytical instruments so as to minimize these temperature or power level variations.
9. Sample pans: It is important to use appropriate pans to contain samples, and to handle them correctly, when carrying out a moisture content analysis. Typically aluminium pans are used because they are relatively cheap and have a high thermal conductivity. These pans usually have lids to prevent spattering of the sample, which would lead to weight loss and therefore erroneous results. Pans should be handled with tongs because fingerprints can contribute to the mass of a sample.
10. Pans should be dried in an oven and stored in a desiccator prior to use to ensure that no residual moisture is attached to them.

D. Advantages and Disadvantages

Advantages: precise; relatively cheap; easy to use; officially sanctioned for many applications;
many samples can be analyzed simultaneously

Disadvantages: Destructive; Unsuitable for some types of food; Time consuming

2. Distillation Methods

Distillation methods are based on *direct* measurement of the amount of water removed from a food sample by evaporation: $\% \text{Moisture} = (M_{\text{WATER}}/M_{\text{INITIAL}}) \times 100$. In contrast, evaporation methods are based on *indirect* measurement of the amount of water removed from a food sample by evaporation: $\% \text{Moisture} = ((M_{\text{INITIAL}} - M_{\text{DRIED}})/M_{\text{INITIAL}}) \times 100$. Basically, distillation methods involve heating a weighed food sample (M_{INITIAL}) in the presence of an organic solvent that is immiscible with water. The water in the sample evaporates and is collected in a graduated glass tube where its mass is determined (M_{WATER}).

A. Dean and Stark Method

Distillation methods are best illustrated by examining a specific example: the Dean and Stark method. A known weight of food is placed in a flask with an organic solvent such as xylene or toluene. The organic solvent must be insoluble with water; have a higher boiling point than water; be less dense than water; and be safe to use. The flask containing the sample and the organic solvent is attached to a condenser by a side arm and the mixture is heated. The water in the sample evaporates and moves up into the condenser where it is cooled and converted back into liquid water, which then trickles into the graduated tube. When no more water is collected in the graduated tube, distillation is stopped and the volume of water is read from the tube.

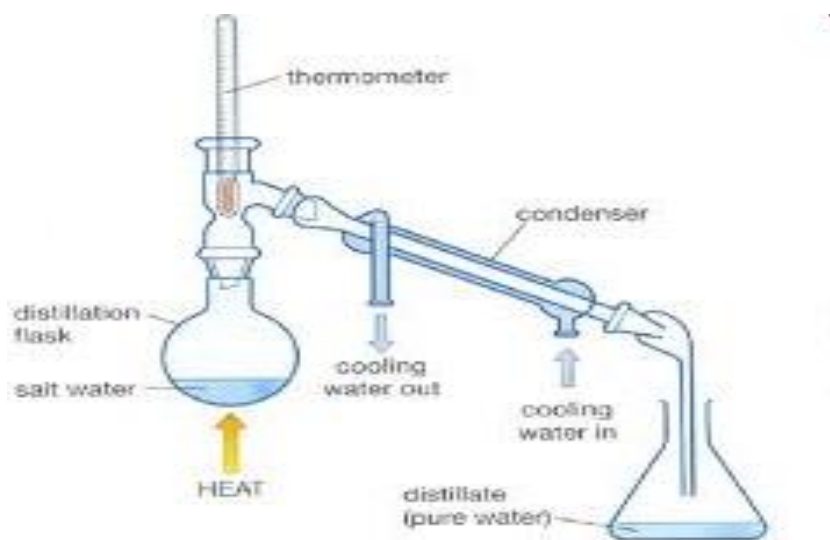


Figure 3.2 Distillation apparatus

I. Practical Considerations

There are a number of practical factors that can lead to erroneous results: (i) emulsions can sometimes form between the water and the solvent which are difficult to separate; (ii) water droplets can adhere to the inside of the glassware, (iii) decomposition of thermally labile samples can occur at the elevated temperatures used.

II. Advantages and Disadvantages

- ➡ Advantages: Suitable for application to foods with low moisture contents; Suitable for application to foods containing volatile oils, such as herbs or spices, since the oils remain dissolved in the organic solvent, and therefore do not interfere with the measurement of the water; Equipment is relatively cheap, easy to setup and operate; Distillation methods have been officially sanctioned for a number of food applications.
- ➡ Disadvantages: Destructive; relatively time-consuming; Involves the use of flammable solvents; Not applicable to some types of foods.

3. Chemical Reaction Methods

Reactions between water and certain chemical reagents can be used as a basis for determining the concentration of moisture in foods. In these methods a chemical reagent is added to the food that reacts specifically with water to produce a measurable change in the properties of the system, *e.g.*, mass, volume, pressure, pH, color, conductivity.

Measurable changes in the system are correlated to the moisture content using calibration curves. To make accurate measurements it is important that the chemical reagent reacts with all of the water molecules present, but not with any of the other components in the food matrix. Two methods that are commonly used in the food industry are the *Karl-Fisher titration* and *gas production* methods. Chemical reaction methods do not usually involve the application of heat and so they are suitable for foods that contain thermally labile substances that would change the mass of the food matrix on heating (*e.g.*, food containing high sugar concentrations) or foods that contain volatile components that might be lost by heating (*e.g.* spices and herbs).

A. Karl-Fisher method

The Karl-Fisher titration is often used for determining the moisture content of foods that have low water contents (*e.g.* dried fruits and vegetables, confectionary, coffee, oils and fats). It is based on the following reaction:



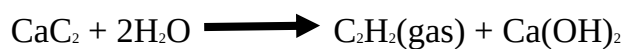
This reaction was originally used because HI is colorless, whereas I₂ is a dark reddish brown color, hence there is a measurable change in color when water reacts with the added chemical reagents. Sulphur dioxide and iodine are gaseous and would normally be lost from solution. For this reason, the above reaction has been modified by adding solvents (*e.g.*, C₅H₅N) that keep the S₂O and I₂ in solution, although the basic principles of the method are the same. The food to be analyzed is placed in a beaker containing solvent and is then titrated with *Karl Fisher reagent* (a solution that contains iodine). While any water remains in the sample the iodine reacts with it and the solution remains colorless (HI), but once all the water has been used up any additional iodine is observed as a dark red brown color (I₂). The volume of iodine solution required to titrate the water is measured and can be related to the moisture content using a pre-prepared calibration curve. The precision of the technique can be improved by using electrical methods to follow the end-point of the reaction, rather than observing a color change. Relatively inexpensive commercial instruments have been developed which are based on the Karl-Fisher titration, and some of these are fully automated to make them less labor intensive.



Figure 3.3 Automated Karl Fischer volumetric titration unit

B. Gas production methods

Commercial instruments are also available that utilize specific reactions between chemical reagents and water that lead to the production of a gas. For example, when a food sample is mixed with powdered calcium carbide the amount of acetylene gas produced is related to the moisture content.



The amount of gas produced can be measured in a number of different ways, including (i) the volume of gas produced, (ii) the decrease in the mass of the sample after the gas is released, and (iii) the increase in pressure of a closed vessel containing the reactants.

4. Physical Methods

A number of analytical methods have been developed to determine the moisture content of foods that are based on the fact that water has appreciably different bulk physical characteristics than the food matrix, *e.g.* density, electrical conductivity or refractive index. These methods are usually only suitable for analysis of foods in which the composition of the food matrix does not change significantly, but the ratio of water-to-food matrix changes. For example, the water content of oil-in-water emulsions can be determined by measuring their density or electrical conductivity because the density and electrical conductivity of water are significantly higher than those of oil. If the composition of the food matrix changes as well as the water content, then it may not be possible to accurately determine the moisture content of the food because more than one food composition may give the same value for the physical property being measured. In these cases, it may be possible to use a combination of two or more physical methods to determine the composition of the food, *e.g.*, density measurements in combination with electrical conductivity measurements.

5. Spectroscopic Methods

Spectroscopic methods utilize the interaction of electromagnetic radiation with materials to obtain information about their composition, *e.g.*, X-rays, UV-visible, NMR, microwaves and IR. The spectroscopic methods developed to measure the moisture content of foods are based on the fact that water absorbs electromagnetic radiation at characteristic wavelengths that are different from the other components in the food matrix. The most widely used physical methods

are based on measurements of the absorption of microwave or infrared energy by foods. Microwave and infrared radiation are absorbed by materials due to their ability to promote the vibration and/or rotation of molecules. The analysis is carried out at a wavelength where the water molecules absorb radiation, but none of the other components in the food matrix do. A measurement of the absorption of radiation at this wavelength can then be used to determine the moisture content: the higher the moisture content, the greater the absorption. Instruments based on this principle are commercially available and can be used to determine the moisture content in a few minutes or less. It is important not to confuse infrared and microwave *absorption* methods with infrared lamp and microwave *evaporation* methods. The former use low energy waves that cause no physical or chemical changes in the food, whereas the latter use high-energy waves to evaporate the water. The major advantage of these methods is that they are capable of rapidly determining the moisture content of a food with little or no sample preparation and are therefore particularly useful for quality control purposes or rapid measurements of many samples.

3.2. Analysis of Ash

The ‘**ash content**’ is a measure of the total amount of minerals present within a food, whereas the ‘**mineral content**’ is a measure of the amount of specific inorganic components present within a food, such as Ca, Na, K, Cl and so on. Determination of the ash and mineral content of foods is important for a number of reasons:

- Nutritional labelling: The concentration and type of minerals present must often be stipulated on the label of a food.
- Quality: The quality of many foods depends on the concentration and type of minerals they contain, including their taste, appearance, texture and stability.
- Microbiological stability: High mineral contents are sometimes used to retard the growth of certain microorganisms.
- Nutrition: Some minerals are essential to a healthy diet (*e.g.*, calcium, phosphorous, potassium and sodium) whereas others can be toxic (*e.g.*, lead, mercury, cadmium and aluminium).
- Processing: It is often important to know the mineral content of foods during processing because this affects the physicochemical properties of foods.

3.2.1 Methods of Ash Content Determination

Ash is the inorganic residue remaining after the water and organic matter have been removed by heating in the presence of oxidizing agents, which provides a measure of the total amount of minerals within a food. Analytical techniques for providing information about the total mineral content are based on the fact that the minerals (the 'analyte') can be distinguished from all the other components (the 'matrix') within a food in some measurable way. The most widely used methods are based on the fact that minerals are not destroyed by heating, and that they have a low volatility compared to other food components. The three main types of analytical procedure used to determine the ash content of foods are based on this principle: dry ashing, wet ashing and low temperature plasma dry ashing. The method chosen for a particular analysis depends on the reason for carrying out the analysis, the type of food analyzed and the equipment available. Ashing may also be used as the first step in preparing samples for analysis of specific minerals, by atomic spectroscopy or the various traditional methods described below. Ash contents of fresh foods rarely exceed 5%, although some processed foods can have ash contents as high as 12%, *e.g.*, dried beef.

3.2.1.1 Sample Preparation

As with all food analysis procedures it is crucial to carefully select a sample whose composition represents that of the food being analyzed and to ensure that its composition does not change significantly prior to analysis. Typically, samples of 2-10g are used in the analysis of ash content. Solid foods are finely ground and then carefully mixed to facilitate the choice of a representative sample. Before carrying out an ash analysis, samples that are high in moisture are often dried to prevent spattering during ashing. High fat samples are usually defatted by solvent extraction, as this facilitates the release of the moisture and prevents spattering. Other possible problems include contamination of samples by minerals in grinders, glassware or crucibles which come into contact with the sample during the analysis. For the same reason, it is recommended to use distilled or deionized water when preparing samples.

1. Dry Ashing

Dry ashing procedures use a high temperature *muffle furnace* capable of maintaining temperatures of between 500 and 600 °C. Water and other volatile materials are vaporized and organic substances are burned in the presence of the oxygen in air to CO₂, H₂O and N₂. Most minerals are converted to oxides, sulfates, phosphates, chlorides or silicates.

Although most minerals have fairly low volatility at these high temperatures, some are volatile and may be partially lost, *e.g.*, iron, lead and mercury. If an analysis is being carried out to determine the concentration of one of these substances then it is advisable to use an alternative ashing method that uses lower temperatures.

The food sample is weighed before and after ashing to determine the concentration of ash present. The food sample is weighed before and after ashing to determine the concentration of ash present. The ash content can be expressed on either a *dry* or *wet* basis:

$$\% \text{ ash (Dry basis)} = \frac{\text{Weight after ashing}}{\text{Weight before ashing}} \times 100 \quad (3.4)$$

$$\% \text{ ash (Wet basis)} = \frac{\text{Weight after ashing}}{\text{Weight before ashing}} \times 100 \quad (3.5)$$

Calculation

Calculate the percentage ash content (wet weight basis) as follows:

$$\% \text{ ash (wet)} = \frac{(\text{Wt. crucible \& ash} - \text{wt. crucible})}{(\text{Weight before ashing (wt. crucible \& sample} - \text{wt. crucible})} \times 100$$

Calculation of ash content on dry basis (when moisture content is known) as follows:

$$\% \text{ ash (Dry)} = \frac{\% \text{ ash (wet)}}{(100 - \% \text{ moisture})} \times 100$$

Where M_{ASH} refers to the mass of the ashed sample, and M_{DRY} and M_{wet} refer to the original masses of the dried and wet samples.

There are a number of different types of crucible available for ashing food samples, including quartz, Pyrex, porcelain, steel and platinum. Selection of an appropriate crucible depends on the sample being analyzed and the furnace temperature used. The most widely used crucibles are made from porcelain because it is relatively inexpensive to purchase, can be used up to high temperatures ($< 1200^{\circ}\text{C}$) and are easy to clean.

Porcelain crucibles are resistant to acids but can be corroded by alkaline samples, and therefore different types of crucible should be used to analyze this type of sample. In addition, porcelain crucibles are prone to cracking if they experience rapid temperature changes. A number of dry ashing methods have been officially recognized for the determination of the ash content of various foods (AOAC Official Methods of Analysis). Typically, a sample is held at 500-600 °C for 24 hours.

- Advantages: Safe, few reagents are required, many samples can be analyzed simultaneously, not labor intensive, and ash can be analyzed for specific mineral content.
- Disadvantages: Long time required (12-24 hours), muffle furnaces are quite costly to run due to electrical costs, loss of volatile minerals at high temperatures, *e.g.*, Cu, Fe, Pb, Hg, Ni, Zn.

Recently, analytical instruments have been developed to dry ash samples based on microwave heating. These devices can be programmed to initially remove most of the moisture (using a relatively low heat) and then convert the sample to ash (using a relatively high heat). Microwave instruments greatly reduce the time required to carry out an ash analysis, with the analysis time often being less than an hour. The major disadvantage is that it is not possible to simultaneously analyze as many samples as in a muffle furnace.

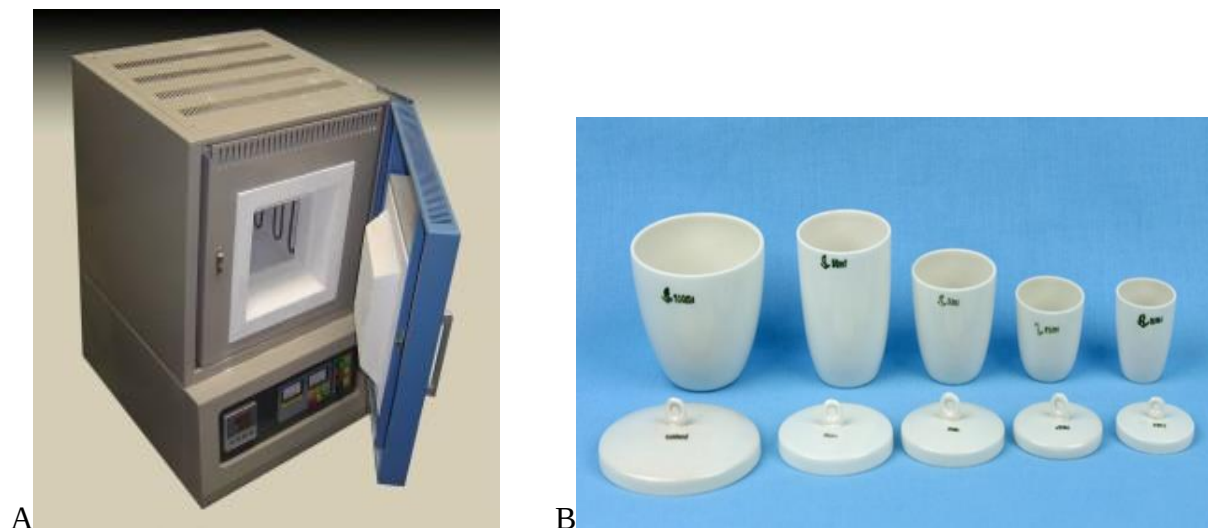


Figure 3.4 A: Muffle furnace B: different size Crucibles

2. Wet Ashing

Wet ashing is primarily used in the preparation of samples for subsequent analysis of specific minerals. It breaks down and removes the organic matrix surrounding the minerals so that they are left in an aqueous solution. A dried ground food sample is usually weighed into a flask containing strong acids and oxidizing agents (*e.g.*, nitric, perchloric and/or sulphuric acids) and then heated. Heating is continued until the organic matter is completely digested, leaving only the mineral oxides in solution. The temperature and time used depends on the type of acids and oxidizing agents used. Typically, a digestion takes from 10 minutes to a few hours at temperatures of about 350°C. The resulting solution can then be analyzed for specific minerals.

- Advantages: Little loss of volatile minerals occurs because of the lower temperatures used, more rapid than dry ashing.
- Disadvantages: Labor intensive, requires a special fume-cupboard if perchloric acid is used because of its hazardous nature, low sample throughput.

3. Low Temperature Plasma Ashing

A sample is placed into a glass chamber which is evacuated using a vacuum pump. A small amount of oxygen is pumped into the chamber and broken down to nascent oxygen ($\text{O}_2 \rightarrow 2\text{O}^\cdot$) by application of an electromagnetic radio frequency field. The organic matter in the sample is rapidly oxidized by the nascent oxygen and the moisture is evaporated because of the elevated temperatures. The relatively cool temperatures ($< 150^\circ\text{C}$) used in low-temperature plasma ashing cause less loss of volatile minerals than other methods.

- Advantages: Less chance of losing trace elements by volatilization
- Disadvantages: Relatively expensive equipment and small sample throughput.

3.2.2. Determination of Water Soluble and Insoluble Ash

As well as the total ash content, it is sometimes useful to determine the ratio of water soluble to water-insoluble ash as this gives a useful indication of the quality of certain foods, *e.g.*, the fruit content of preserves and jellies. Ash is diluted with distilled water then heated to nearly boiling, and the resulting solution is filtered.

The amount of soluble ash is determined by drying the filtrate, and the insoluble ash is determined by rinsing, drying and ashing the filter paper.

3.2.3. Comparison of Ashing Methods

The conventional dry ashing procedure is simple to carry out, is not labor intensive, requires no expensive chemicals and can be used to analyze many samples simultaneously. Nevertheless, the procedure is time-consuming and volatile minerals may be lost at the high temperatures used. Microwave instruments are capable of speeding up the process of dry ashing. Wet ashing and low temperature plasma ashing are more rapid and cause less loss of volatile minerals because samples are heated to lower temperatures. Nevertheless, the wet ashing procedure requires the use of hazardous chemicals and is labor intensive, while the plasma method requires expensive equipment and has a low sample throughput.

3.3. Determination of Specific Mineral Content

Knowledge of the concentration and type of specific minerals present in food products is often important in the food industry. The major physicochemical characteristics of minerals that are used to distinguish them from the surrounding matrix are: their low volatility; their ability to react with specific chemical reagents to give measurable changes; and their unique electromagnetic spectra. The most effective means of determining the type and concentration of specific minerals in foods is to use atomic absorption or emission spectroscopy. Instruments based on this principle can be used to quantify the entire range of minerals in foods, often to concentrations as low as a few ppm. For these reasons they have largely replaced traditional methods of mineral analysis in institutions that can afford to purchase and maintain one, or that routinely analyze large numbers of samples. Institutions that do not have the resources or sample throughput to warrant purchasing an atomic spectroscopy instrument rely on more traditional methods that require chemicals and equipment commonly found in food laboratories. Many of the minerals of importance to food scientists can be measured using one of these traditional methods.

3.2.5.1 Sample preparation

Many of the analytical methods used to determine the specific mineral content of foods require that the minerals be dissolved in an aqueous solution. For this reason, it is often necessary to isolate the minerals from the organic matrix surrounding them prior to the analysis. This is usually carried out by ashing a sample using one of the methods described in the previous section. It is important that the ashing procedure does not alter the mineral concentration in the food due to volatilization.

Another potential source of error in mineral analysis is the presence of contaminants in the water, reagents or glassware. For this reason, ultrapure water or reagents should be used, and/or a blank should be run at the same time as the sample being analyzed. A blank uses the same glassware and reagents as the sample being analyzed and therefore should contain the same concentration of any contaminants. The concentration of minerals in the blank is then subtracted from the value determined for the sample. Some substances can interfere with analysis of certain minerals, and should therefore be eliminated prior to the analysis or accounted for in the data interpretation. The principles of a number of the most important methods for analyzing minerals are described below. Many more traditional methods can be found in the AOAC Official Methods of Analysis.

3.2.5.2 Gravimetric Analysis

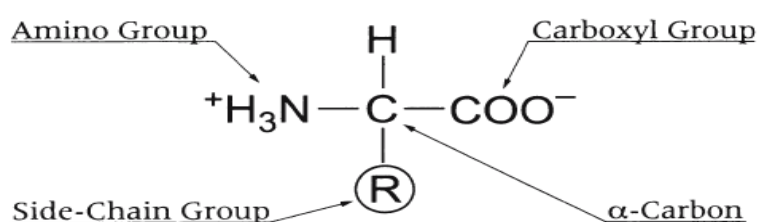
The element (mineral) to be analyzed is precipitated from solution by adding a reagent that reacts with it to form an insoluble complex with a known chemical formula. The precipitate is separated from the solution by filtration, rinsed, dried and weighed. The amount of mineral present in the original sample is determined from knowledge of the chemical formula of the precipitate. For example, the amount of chloride in a solution can be determined by adding excess silver ions to form an insoluble silver chloride precipitate, because it is known that Cl is 24.74% of AgCl. Gravimetric procedures are only suitable for large food samples, which have relatively high concentrations of the mineral being analyzed. They are not suitable for analysis of trace elements because balances are not sensitive enough to accurately weigh the small amount of precipitate formed.

3.2.5.3 Colorimetric methods

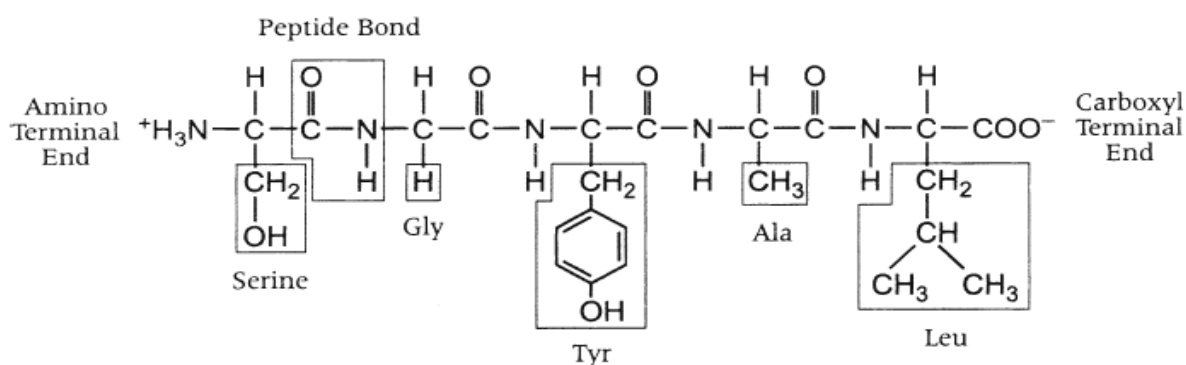
These methods rely on a change in color of a reagent when it reacts with a specific mineral in solution which can be quantified by measuring the absorbance of the solution at a specific wavelength using a spectrophotometer. Colorimetric methods are used to determine the concentration of a wide variety of different minerals. Vandate is often used as a colorimetric reagent because it changes color when it reacts with minerals. For example, the phosphorous content of a sample can be determined by adding a vandate-molybdate reagent to the sample. This forms a colored complex (yellow-orange) with the phosphorous which can be quantified by measuring the absorbance of the solution at 420nm, and comparing with a calibration curve. Different reagents are also available to colorimetrically determine the concentration of other minerals.

3.3. Analysis of Proteins

Proteins are polymers of amino acids. They are composed of elements including hydrogen, carbon, nitrogen, oxygen, and sulphur. Twenty α -amino acids are the building blocks of proteins; the amino acid residues in a protein are linked by peptide bonds. Nitrogen is the most distinguishing element present in proteins. Twenty different types of amino acids occur naturally in proteins. Proteins differ from each other according to the type, number and sequence of amino acids that make up the polypeptide backbone. As a result they have different molecular structures, nutritional attributes and physiochemical properties. The structure of amino acid and protein respectively are shown bellow.



Amino acid structure



Protein structure

Proteins are important constituents of foods for a number of different reasons. They are a major source of *energy*, as well as containing essential amino-acids, such as lysine, tryptophan, methionine, leucine, isoleucine and valine, which are essential to human health, but which the body cannot synthesize. Proteins are also the major structural components of many natural foods, often determining their overall texture, *e.g.*, tenderness of meat or fish products. Isolated proteins are often used in foods as ingredients because of their unique functional properties, *i.e.*, their ability to provide desirable appearance, texture or stability. Typically, proteins are used as gelling agents, emulsifiers, foaming agents and thickeners. Many food proteins are enzymes which are capable of enhancing the rate of certain biochemical reactions. These reactions can have either a favorable or detrimental effect on the overall properties of foods.

Food analysts are interested in knowing the total concentration, type, molecular structure and functional properties of the proteins in foods.

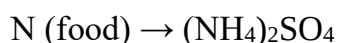
3.3.1. Methods of Total Protein Concentration Determination

3.1.1.1. Kjeldahl method

The Kjeldahl method was developed in 1883 by a brewer called Johann Kjeldahl. A food is digested with a strong acid so that it releases nitrogen which can be determined by a suitable titration technique. The amount of protein present is then calculated from the nitrogen concentration of the food. The same basic approach is still used today, although a number of improvements have been made to speed up the process and to obtain more accurate measurements. It is usually considered to be *the* standard method of determining protein concentration. Because the Kjeldahl method does not measure the protein content directly a *conversion factor (F)* is needed to convert the measured nitrogen concentration to a protein concentration. A conversion factor of 6.25 (equivalent to 0.16 g nitrogen per gram of protein) is used for many applications, however, this is only an average value, and each protein has a different conversion factor depending on its amino-acid composition. The Kjeldahl method can conveniently be divided into three steps: digestion, neutralization and titration.

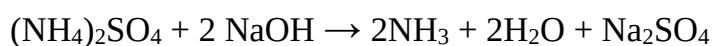
I. Digestion

The food sample to be analyzed is weighed into a *digestion flask* and then digested by heating it in the presence of sulphuric acid (an oxidizing agent which digests the food), anhydrous sodium sulphate (to speed up the reaction by raising the boiling point) and a catalyst, such as copper, selenium, titanium, or mercury (to speed up the reaction). Digestion converts any nitrogen in the food (other than that which is in the form of nitrates or nitrites) into ammonia, and other organic matter to CO₂ and H₂O. Ammonia gas is not liberated in an acid solution because the ammonia is in the form of the ammonium ion (NH₄⁺) which binds to the sulphate ion (SO₄²⁻) and thus remains in solution:



II. Distillation (Neutralization)

After the digestion has been completed the digestion flask is connected to a *receiving flask* by a tube. The solution in the digestion flask is then made alkaline by addition of sodium hydroxide, which converts the ammonium sulfate into ammonia gas:



The ammonia gas that is formed is liberated from the solution and moves out of the digestion flask and into the receiving flask - which contains an excess of boric acid. The low pH of the solution in the receiving flask converts the ammonia gas into the ammonium ion, and simultaneously converts the boric acid to the borate ion:

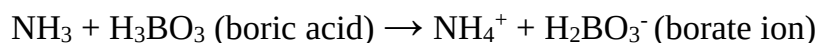
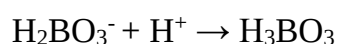


Figure 3.8 A: Kjeldahl Digestion block B: Kjeldahl distillation unit

III. Titration

The nitrogen content is then estimated by titration of the ammonium borate formed with standard sulfuric or hydrochloric acid, using a suitable indicator to determine the end-point of the reaction.



The concentration of hydrogen ions (in moles) required to reach the end-point is equivalent to the concentration of nitrogen that was in the original food (Equation 3). The following equation can be used to determine the nitrogen concentration of a sample that weighs m grams using a x M HCl acid solution for the titration:

$$\%N = \frac{V \text{ HCl(L)} \times N \text{ HCl} \times 14}{\text{sample wt.}} \times 100 \quad \text{----- (3.5)}$$

Or

$$\%N = \frac{V_{HCl(L)}(V_s - V_b) \times N_{HCl} \times 14}{\dots} \times 100 \quad (3.6)$$

Where v_s and v_b are the titration volumes of the sample and blank, and 14g is the molecular weight of nitrogen N. A blank sample is usually ran at the same time as the material being analyzed to take into account any residual nitrogen which may be in the reagents used to

$$\% \text{Protein} = F \times \%N$$

carry out the analysis. Once the nitrogen content has been determined it is converted to a protein content using the appropriate conversion factor:

$$\text{---} \quad (3.7)$$

Table 3.2 Nitrogen to Protein conversion factors for some Foods

Foods	factors
Egg or meat	6.25
Dairy products	6.38
Wheat	5.70
Other cereal grains and oilseeds	6.25
Almonds	5.18
Peanuts	5.46
Other tree nuts and coconut	5.30

Advantages and Disadvantages

Advantages: The Kjeldahl method is widely used internationally and is still the standard method for comparison against all other methods. Its universality, high precision and good reproducibility have made it the major method for the estimation of protein in foods.

Disadvantages: It does not give a measure of the true protein, since all nitrogen in foods is not in the form of protein. Different proteins need different correction factors because they have different amino acid sequences. The use of concentrated sulphuric acid at high temperatures

poses a considerable hazard, as does the use of some of the possible catalysts. The technique is time consuming to carry-out.

3.4 Analysis of Lipids

Lipids are one of the major constituents of foods, and are important in our diet for a number of reasons. They are a major source of energy and provide essential lipid nutrients. Nevertheless, over-consumption of certain lipid components can be detrimental to our health, *e.g.* cholesterol and saturated fats. In many foods the lipid component plays a major role in determining the overall physical characteristics, such as flavor, texture, mouthfeel and appearance. For this reason, it is difficult to develop low-fat alternatives of many foods; because once the fat is removed some of the most important physical characteristics are lost. Finally, many fats are prone to lipid oxidation, which leads to the formation of off-flavors and potentially harmful products. Some of the most important properties of concern to the food analyst are:

- Total lipid concentration
- Type of lipids present
- Physicochemical properties of lipids, *e.g.*, crystallization, melting point, smoke point, rheology, density and color
- Structural organization of lipids within a food

3.4.1 Properties of Lipids in Foods

Lipids are usually defined as those components that are soluble in organic solvents (such as ether, hexane or chloroform), but are insoluble in water. This group of substances includes triacylglycerols, diacylglycerols, monoacylglycerols, free fatty acids, phospholipids, sterols, carotenoids and vitamins A and D. The lipid fraction of a fatty food therefore contains a complex mixture of different types of molecule. Even so, triacylglycerols are the major component of most foods, typically making up more than 95 to 99% of the total lipids present. Triacylglycerols are esters of three fatty acids and a glycerol molecule. The fatty acids normally found in foods vary in chain length, degree of unsaturation and position on the glycerol molecule.

Consequently, the triacylglycerol fraction itself consists of a complex mixture of different types of molecules. Each type of fat has a different profile of lipids present which determines the precise nature of its nutritional and physiochemical properties. The terms fat, oil and lipid are often used interchangeably by food scientists. Although sometimes the term *fat* is used to

describe those lipids that are solid at the specified temperature, whereas the term *oil* is used to describe those lipids that are liquid at the specified temperature.

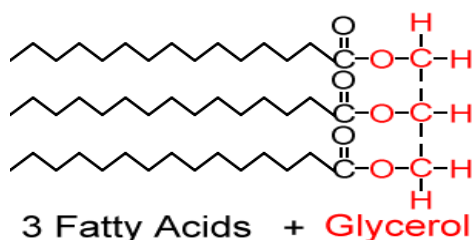


Figure 3.10 structure of Triglycerol

3.4.2 Sample Selection and Preservation

As with any analytical procedure, the validity of the results depends on proper sampling and preservation of the sample prior to analysis. Ideally, the composition of the sample analyzed should represent as closely as possible that of the food from which it was taken. The sample preparation required in lipid analysis depends on the type of food being analyzed (*e.g.* meat, milk, margarine, cookie, dairy cream), the nature of the lipid component (*e.g.* volatility, susceptibility to oxidation, physical state) and the type of analytical procedure used (*e.g.* solvent extraction, non-solvent extraction or instrumental). In order, to decide the most appropriate sample preparation procedure it is necessary to have knowledge of the physical structure and location of the principal lipids present in the food. Since each food is different it is necessary to use different procedures for each one. Official methods have been developed for specific types of foods that stipulate the precise sample preparation procedure that should be followed. In general, sample preparation should be carried out using an environment that minimizes any changes in the properties of the lipid fraction. If lipid oxidation is a problem it is important to preserve the sample by using a nitrogen atmosphere, cold temperature, low light or adding antioxidants. If the solid fat content or crystal structure is important it may be necessary to carefully control the temperature and handling of the sample.

3.4.3 Methods of Total Lipid Concentration Determination

It is important to be able to accurately determine the total fat content of foods for a number of reasons:

- Economic (not to give away expensive ingredients)
- Legal (to conform to standards of identity and nutritional labelling laws)
- Health (development of low fat foods)
- Quality (food properties depend on the total lipid content)
- Processing (processing conditions depend on the total lipid content)

The principle physicochemical characteristics of lipids (the "analyte") used to distinguish them from the other components in foods (the "matrix") are their solubility in organic solvents, immiscibility with water, physical characteristics (*e.g.*, relatively low density) and spectroscopic properties. The analytical techniques based on these principles can be conveniently categorized into three different types: (i) solvent extraction; (ii) non-solvent extraction and (iii) instrumental methods.

3.4.3.1. Solvent Extraction

The fact that lipids are soluble in organic solvents, but insoluble in water, provides the food analyst with a convenient method of separating the lipid components in foods from water soluble components, such as proteins, carbohydrates and minerals. In fact, solvent extraction techniques are one of the most commonly used methods of isolating lipids from foods and of determining the total lipid content of foods.

Sample Preparation

The preparation of a sample for solvent extraction usually involves a number of steps:

- ➡ Drying sample. It is often necessary to dry samples prior to solvent extraction, because many organic solvents cannot easily penetrate into foods containing water, and therefore extraction would be inefficient.
- ➡ Particle size reduction. Dried samples are usually finely ground prior to solvent extraction to produce a more homogeneous sample and to increase the surface area of lipid exposed to the solvent. Grinding is often carried out at low temperatures to reduce the tendency for lipid oxidation to occur.

- ➡ Acid hydrolysis: Some foods contain lipids that are complexed with proteins (lipoproteins) or polysaccharides (glycolipids). To determine the concentration of these components it is necessary to break the bonds which hold the lipid and non-lipid components together prior to solvent extraction. Acid hydrolysis is commonly used to release bound lipids into easily extractable forms, *e.g.* a sample is digested by heating it for 1 hour in the presence of 3N HCl acid.
- ➡ Solvent Selection: The *ideal* solvent for lipid extraction would completely extract all the lipid components from a food, while leaving all the other components behind. In practice, the efficiency of solvent extraction depends on the *polarity* of the lipids present compared to the polarity of the solvent. Polar lipids (such as glycolipids or phospholipids) are more soluble in polar solvents (such as alcohols), than in non-polar solvents (such as hexane). On the other hand, non-polar lipids (such as triacylglycerols) are more soluble in non-polar solvents than in polar ones. The fact that different lipids have different polarities means that it is impossible to select a single organic solvent to extract them all. Thus the total lipid content determined by solvent extraction depends on the nature of the organic solvent used to carry out the extraction: the total lipid content determined using one solvent may be different from that determined using another solvent. In addition to the above considerations, a solvent should also be inexpensive, have a relatively low boiling point (so that it can easily be removed by evaporation), be non-toxic and be nonflammable (for safety reasons). It is difficult to find a single solvent which meets all of these requirements. Ethyl ether and petroleum ether are the most commonly used solvents, but pentane and hexane are also used for some foods.

A. Batch Solvent Extraction

These methods are based on mixing the sample and the solvent in a suitable container, *e.g.*, a separatory funnel. The container is shaken vigorously and the organic solvent and aqueous phase are allowed to separate (either by gravity or centrifugation). The aqueous phase is then decanted off, and the concentration of lipid in the solvent is determined by evaporating the solvent and measuring the mass of lipid remaining:

$$\% \text{Lipid} = (M_{\text{lipid}} / M_{\text{sample}}) \times 100$$

B. Semi-Continuous Solvent Extraction

Semi-continuous solvent extraction methods are commonly used to increase the efficiency of lipid extraction from foods. The Soxhlet method is the most commonly used example of a semi-continuous method. In the Soxhlet method a sample is dried, ground into small particles and placed in a porous thimble. The thimble is placed in an extraction chamber, which is suspended above a flask containing the solvent and below a condenser. The flask is heated and the solvent evaporates and moves up into the condenser where it is converted into a liquid that trickles into the extraction chamber containing the sample. Eventually, the solvent builds up in the extraction chamber and completely surrounds the sample. The extraction chamber is designed so that when the solvent surrounding the sample exceeds a certain level it overflows and trickles back down into the boiling flask. As the solvent passes through the sample it extracts the lipids and carries them into the flask. The lipids then remain in the flask because of their low volatility. At the end of the extraction process, which typically lasts a few hours, the flask containing the solvent and lipid is removed, the solvent is evaporated and the mass of lipid remaining is measured (M_{lipid}). The percentage of lipid in the initial sample (M_{sample}) can then be calculated:

$$\% \text{ Lipid} = (M_{\text{lipid}}/M_{\text{sample}}) \times 100$$

A number of instrument manufacturers have designed modified versions of the Soxhlet method that can be used to determine the total lipid content more easily and rapidly.

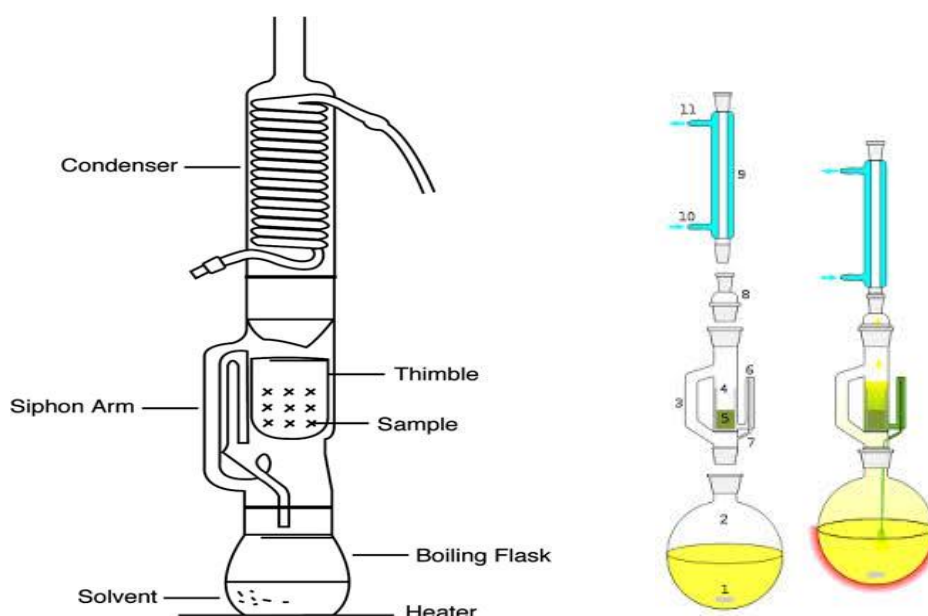


Figure 3.11 Soxhlet apparatus

C. Continuous Solvent Extraction

The Goldfish method is similar to the Soxhlet method except that the extraction chamber is designed so that the solvent just trickles through the sample rather than building up around it. This reduces the amount of time required to carry out the extraction, but it has the disadvantage that *channelling* of the solvent can occur, *i.e.*, the solvent may preferentially take certain routes through the sample and therefore the extraction is inefficient. This is not a problem in the Soxhlet method because the sample is always surrounded by solvent.



Figure 3.12 The Goldfish apparatus

3.4.3.2. Non-solvent Liquid Extraction Methods

A number of liquid extraction methods do not rely on organic solvents, but use other chemicals to separate the lipids from the rest of the food. The Babcock, Gerber and Detergent methods are examples of non-solvent liquid extraction methods for determining the lipid content of milk and some other dairy products.

A. Babcock Method

A specified amount of milk is accurately pipetted into a specially designed flask (the Babcock bottle). Sulphuric acid is mixed with the milk, which digests the protein, generates heat, and breaks down the fat globule membrane that surrounds the droplets, thereby releasing the fat. The sample is then centrifuged while it is hot (55-60°C) which causes the liquid fat to rise into the neck of the Babcock bottle.

The neck is graduated to give the amount of milk fat present in wt%. The Babcock method takes about 45 minutes to carry out, and is precise to within 0.1%. It does not determine phospholipids in milk, because they are located in the aqueous phase or at the boundary between the lipid and aqueous phases.

B. Gerber Method

This method is similar to the Babcock method except that a mixture of sulphuric acid and *isoamyl alcohol*, and a slightly different shaped bottle, are used. It is faster and simpler to carry out than the Babcock method. Sulphuric acid is used to break the bond b/n the fat and protein and other constituents while isoamyl alcohol is for clear separation of the fat from the rest parts of the sample. The isoamyl alcohol is used to prevent charring of the sugars by heat and sulphuric acid which can be a problem in the Babcock method since it makes it difficult to read the fat content from the graduated flask. This method is used mainly in Europe, whilst the Babcock method is used mainly in the USA. As with the Babcock method, it does not determine phospholipids. In this method the sample, sulphuric acid and isoamyl alcohol is added to the butyrometer and centrifuged by using Gerber centrifuge. The fat content is then directly read from the butyrometer.



Figure 3.13 Butyrometer

C. Detergent Method

This method was developed to overcome the inconvenience and safety concerns associated with the use of highly corrosive acids. A sample is mixed with a combination of surfactants in a Babcock bottle. The surfactants displace the fat globule membrane which surrounds the emulsion droplets in milk and causes them to coalesce and separate. The sample is centrifuged which allows the fat to move into the graduated neck of the bottle, where its concentration can then be determined.

3.4.3.3. Instrumental methods

There are a wide variety of different instrumental methods available for determining the total lipid content of food materials. These can be divided into three different categories according to their physicochemical principles: (i) measurement of bulk physical properties, (ii) measurement of adsorption of radiation, and (iii) measurement of scattering of radiation. Each instrumental method has its own advantages and disadvantages, and range of foods to which it can be applied.

A. Measurement of bulk physical properties

- **Density:** The density of liquid oil is less than that of most other food components, and so there is a decrease in density of a food as its fat content increases. Thus the lipid content of foods can be determined by measuring their density.
- **Electrical conductivity:** The electrical conductivity of lipids is much smaller than that of aqueous substances, and so the conductivity of a food decreases as the lipid concentration increases. Measurements of the overall electrical conductivity of foods can therefore be used to determine fat contents.
- **Ultrasonic velocity:** The speed at which an ultrasonic wave travels through a material depends on the concentration of fat in a food. Thus the lipid content can be determined by measuring its ultrasonic velocity. This technique is capable of rapid, nondestructive on-line measurements of lipid content.

3.5. Analysis of Carbohydrates

Carbohydrates are one of the most important components in many foods. Carbohydrates may be present as isolated molecules or they may be physically associated or chemically bound to other molecules. Individual molecules can be classified according to the number of monomers that they contain as monosaccharides, oligosaccharides or polysaccharides. Molecules in which the carbohydrates are covalently attached to proteins are known as *glycoproteins*, whereas those in which the carbohydrates are covalently attached to lipids are known as *glycolipids*. Some carbohydrates are digestible by humans and therefore provide an important source of energy, whereas others are indigestible and therefore do not provide energy. Indigestible carbohydrates form part of a group of substances known as *dietary fiber*, which also includes lignin. Consumption of significant quantities of dietary fiber has been shown to be beneficial to human nutrition, helping reduce the risk of certain types of cancer, coronary heart disease, diabetes and constipation. As well as being an important source of energy and dietary fiber, carbohydrates also contribute to the sweetness, appearance and textural characteristics of many foods. It is important to determine the type and concentration of carbohydrates in foods for a number of reasons.

- Standards of Identity - foods must have compositions which conform to government regulations
- Nutritional Labeling - to inform consumers of the nutritional content of foods
- Detection of Adulteration - each food type has a carbohydrate "fingerprint"
- Food Quality - physicochemical properties of foods such as sweetness, appearance, stability and texture depend on the type and concentration of carbohydrates present.
- Economic - industry doesn't want to give away expensive ingredients
- Food Processing - the efficiency of many food processing operations depends on the type and concentration of carbohydrates that are present

3.5.2 Methods of Analysis

A large number of analytical techniques have been developed to measure the total concentration and type of carbohydrates present in foods (see *Food Analysis* by Nielsen for more details). The carbohydrate content of a food can be determined by calculating the percent remaining after all the other components have been measured:

$$\% \text{carbohydrates} = 100 - \% \text{moisture} - \% \text{protein} - \% \text{lipid} - \% \text{mineral}$$

Nevertheless, this method can lead to erroneous results due to experimental errors in any of the other methods, and so it is usually better to directly measure the carbohydrate content for accurate measurements.

3. Density

The density of a material is its mass divided by its volume. The density of aqueous solutions increases as the carbohydrate concentration increases. Thus the carbohydrate concentration can be determined by measuring density, *e.g.*, using density bottles or hydrometers.

This technique is routinely used in industry for determination of carbohydrate concentrations of juices and beverages.



Figure 3.15 Hydrometer

3.9 Study Questions

1. Identify five factors that one would need to consider when choosing a moisture analysis method for a specific food product.
2. Why is standardized methodology needed for moisture determinations?
3. What are the potential advantages of using a vacuum oven rather than a forced draft oven for moisture content determination?
4. A new instrument based on infrared principles has been received in your laboratory to be used in moisture analysis. Briefly describe the way you would ascertain if the new instrument would meet your satisfaction and company standards.
5. You are determining the total ash content of a product using the convectional dry ashing method. Your boss asks you to switch to a convectional wet ashing method because he/she has heard it takes less time than dry ashing.
 - (a) Do you agree or disagree with your boss concerning the time issue, and why?
 - (b) Not considering the time issues, why might you want to continue using dry ashing, *and* why might you change to wet ashing?

6. How would you recommend to your technician to overcome the following problems that could arise in conventional dry ashing of various foods?
- (a) You seem to be getting volatilization of phosphorus, when you want to later determine the phosphorus content.
 - (b) You are getting incomplete combustion of a product high in sugar after a typical dry ashing procedure (i.e., the ash is dark colored, not white or pale gray).
 - (c) The typical procedure takes too long for your purpose. You need to speed up the procedure, but you do not want to use the standard wet ashing procedure.
7. Identify an advantage and disadvantage of using microwave wet digesters or microwave muffle furnaces compared with convectional units.
8. A grain was found to contain 11.5% moisture. A 5.2146- g sample was placed into a crucible (28.5053 g). The ashed crucible weighed 28.5939 g. Calculate the percentage ash on:
- (a) as-received (wet weight) basis and
 - (b) dry matter basis.
9. A dehydrated precooked bean was analyzed for crude protein content in duplicate using the Kjeldahl method. The following data were recorded:
- Moisture content = 8.00%
 - Wt of Sample 1 = 1.015 g
 - Wt of Sample 2 = 1.025 g
 - Normality of HCl used for titration = 0.1142N
 - HCl used for Sample 1 = 22.0 ml
 - HCl used for Sample 2 = 22.5 ml
 - HCl used for reagent blank = 0.2 ml.
- (b) Calculate crude protein content on both wet and dry weight basis of the bean, assuming bean protein contains 17.5% nitrogen.

